

# Technical specification

Valve-regulated lead-acid battery with Absorbent Glass Mat (AGM) separator. Sealed construction for standby power applications.

**12 V**

6 cells

**5 Ah**

20 h to 10.5 V at 25°C

**22 W/cell**

15 min to 1.67 V/cell at 25°C

## Electrical performance

|                       |                                   |
|-----------------------|-----------------------------------|
| 20-hour capacity      | 5.00 Ah at 0.25 A to 10.5 V, 25°C |
| 10-hour capacity      | 4.70 Ah at 0.47 A to 10.5 V, 25°C |
| 1-hour capacity       | 2.60 Ah at 2.60 A to 9.6 V, 25°C  |
| Internal resistance   | $\leq 35\text{m}\Omega$ at 25°C   |
| Maximum discharge     | 75A for 5 seconds at 25°C         |
| Short-circuit current | 270A                              |
| Self-discharge        | 3% average per month at 20°C      |
| Design life / standby | 5 years                           |

## Charging and operating conditions

|                          |                                                            |
|--------------------------|------------------------------------------------------------|
| Charge method            | Constant voltage                                           |
| Float / cycle voltage    | 13.38-13.80 V / 14.40-14.70 V at 25°C                      |
| Maximum charge current   | 2.00A                                                      |
| Temperature compensation | Float: -20 mV/°C per battery; cycle: -30 mV/°C per battery |
| Operating temperatures   | Discharge: -20 to +60°C; charge: -10 to +60°C              |
| Storage temperature      | -20 to +60°C                                               |

## Mechanical specification

|                        |                                         |
|------------------------|-----------------------------------------|
| Dimensions (L × W × H) | 90 × 70 × 101 mm; overall height 107 mm |
| Approximate weight     | 1.72 kg $\pm 5\%$                       |
| Terminal / torque      | See terminal drawing; Not specified     |
| Container              | ABS                                     |

# Construction & applications

High-output AGM battery for standby power applications. Lead-calcium-tin plates, optimized grids and paste support high energy density. Rated power: 22 W per cell for 15 minutes to 1.67 V/cell at 25°C; design life: 5 years.

## Battery construction

|                            |                                 |
|----------------------------|---------------------------------|
| Positive / negative plates | Lead-calcium-tin alloy          |
| Grid design                | Optimized for high power output |
| Plate paste                | Optimized formulation           |
| Internal pressure          | Low-pressure operation          |
| Gas management             | Gas recombination               |
| Charging modes             | Cycle and standby use           |

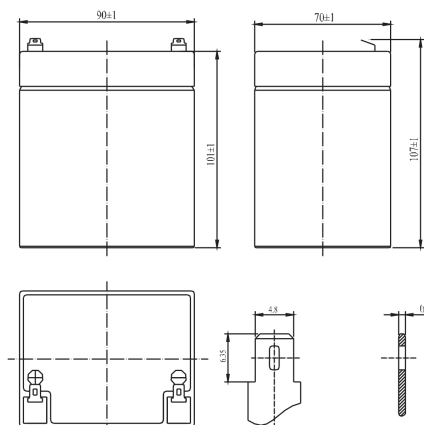
## Features

- Lead-calcium-tin plate alloy.
- High energy density and power output.
- Low internal pressure; gas recombination.
- Application-specific grid and paste design.
- Sealed, valve-regulated AGM construction.
- Shelf life without recharge: 6 months at 20°C.

## Applications

- Uninterruptible power supplies (UPS).
- Telecommunications backup power.
- Power and energy infrastructure.
- Security and fire alarm systems.
- Renewable-energy systems.
- Medical equipment.
- Emergency lighting.
- Data centres.

## Dimensions and terminal details / mm



Dimensions and tolerances govern; do not scale the drawing.

# Discharge performance

Fully charged battery / 25°C

## Constant-current discharge / A

| V/cell | 3m    | 5m    | 10m   | 15m   | 20m   | 30m  | 45m  | 1h   | 1.5h | 2h   | 3h   | 4h   | 5h   | 6h   | 10h  | 20h  |
|--------|-------|-------|-------|-------|-------|------|------|------|------|------|------|------|------|------|------|------|
| 1.60   | 35.01 | 27.00 | 19.60 | 13.60 | 10.60 | 8.00 | 5.50 | 3.98 | 3.30 | 2.60 | 1.23 | 1.11 | 0.98 | 0.86 | 0.51 | 0.28 |
| 1.67   | 33.28 | 25.50 | 18.70 | 12.30 | 9.30  | 7.65 | 4.87 | 3.68 | 3.00 | 2.30 | 1.01 | 0.99 | 0.96 | 0.94 | 0.49 | 0.27 |
| 1.70   | 31.65 | 23.30 | 17.60 | 11.50 | 8.90  | 7.45 | 4.32 | 3.44 | 2.80 | 2.20 | 0.96 | 0.94 | 0.92 | 0.90 | 0.48 | 0.27 |
| 1.75   | 30.25 | 22.12 | 16.50 | 10.60 | 8.50  | 7.32 | 3.86 | 3.12 | 2.60 | 2.00 | 0.92 | 0.91 | 0.89 | 0.88 | 0.47 | 0.26 |
| 1.80   | 29.89 | 21.08 | 15.30 | 9.80  | 7.60  | 6.89 | 3.52 | 2.89 | 2.40 | 1.90 | 0.88 | 0.86 | 0.84 | 0.82 | 0.46 | 0.25 |

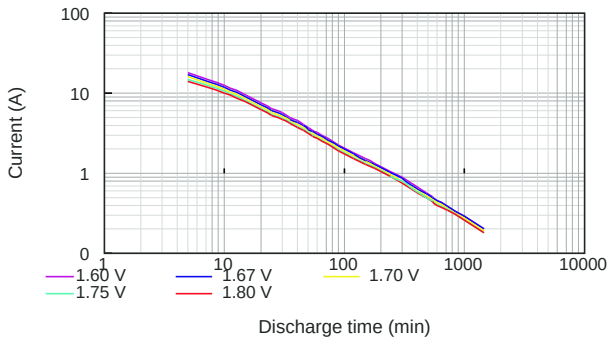
## Constant-power discharge / W per cell

| V/cell | 3m    | 5m    | 10m   | 15m   | 20m   | 30m   | 45m  | 1h   | 1.5h | 2h   | 3h   | 4h   | 5h   | 6h   | 10h  | 20h  |
|--------|-------|-------|-------|-------|-------|-------|------|------|------|------|------|------|------|------|------|------|
| 1.60   | 55.80 | 49.50 | 31.60 | 23.50 | 18.70 | 13.80 | 9.17 | 7.10 | 6.00 | 4.95 | 2.80 | 2.50 | 2.10 | 1.75 | 1.05 | 0.46 |
| 1.67   | 51.20 | 47.10 | 29.10 | 22.00 | 17.40 | 12.80 | 8.20 | 6.99 | 5.90 | 4.75 | 2.51 | 2.20 | 1.98 | 1.63 | 0.99 | 0.45 |
| 1.70   | 46.50 | 44.80 | 27.40 | 20.50 | 16.20 | 11.80 | 8.01 | 6.45 | 5.40 | 4.34 | 2.23 | 2.00 | 1.67 | 1.58 | 0.84 | 0.45 |
| 1.75   | 44.10 | 42.30 | 25.50 | 18.90 | 14.60 | 10.34 | 7.98 | 6.32 | 5.30 | 4.24 | 2.16 | 1.80 | 1.43 | 1.39 | 0.72 | 0.43 |
| 1.80   | 42.60 | 39.80 | 23.10 | 16.50 | 13.20 | 9.87  | 7.65 | 6.10 | 5.10 | 4.09 | 2.08 | 1.70 | 1.24 | 1.21 | 0.62 | 0.42 |

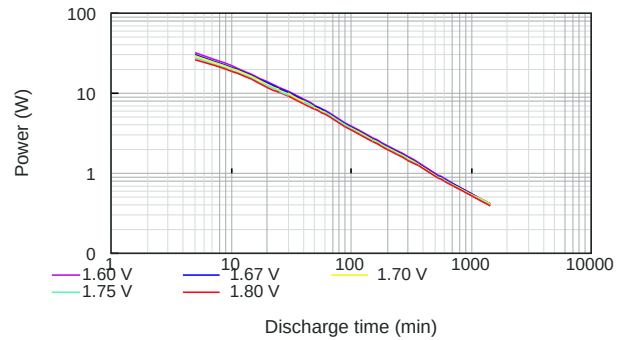
End voltage is per cell. Multiply W/cell by 6 for power per 12 V battery. Current is not divided by 6. Values are averages obtained within three charge/discharge cycles, not guaranteed minimums. Fully charge before testing. Product information may change without notice.

### Characteristic curves

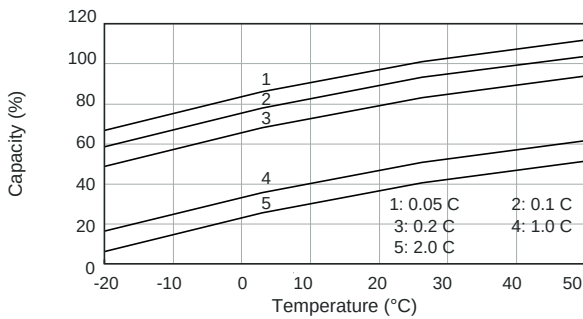
#### Constant-current discharge



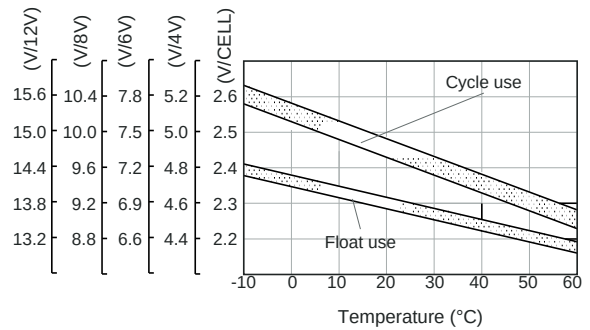
#### Constant-power discharge



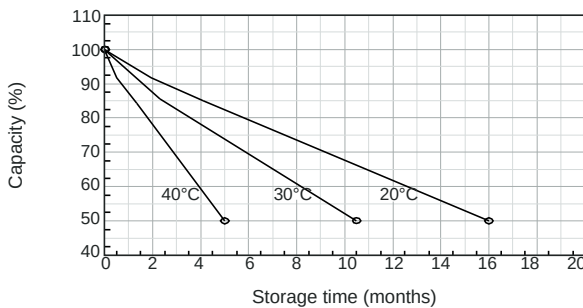
#### Capacity vs. temperature



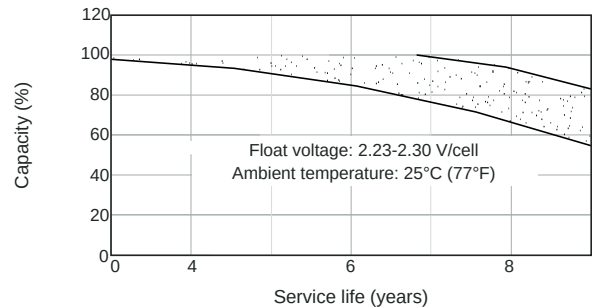
#### Charge voltage vs. temperature



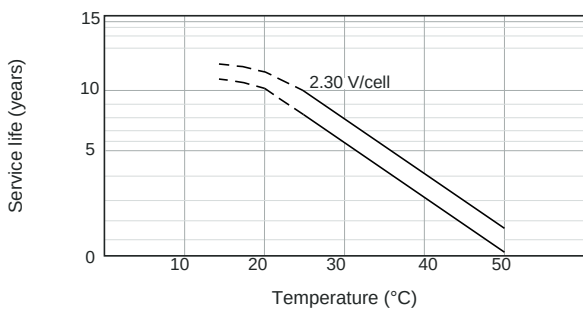
#### Self-discharge



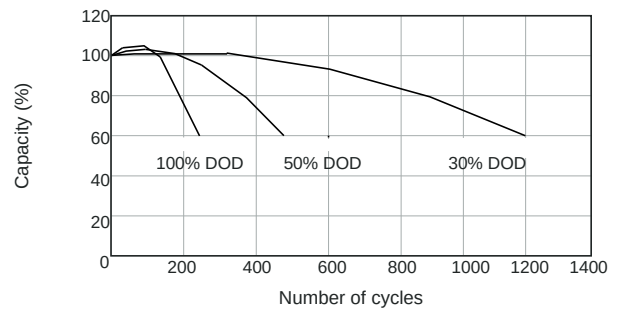
#### Standby service life



#### Float life vs. temperature



#### Cycle life vs. depth of discharge



Typical characteristic curves. Model-specific ratings are given on pages 1 and 3. Design life is not a warranty.